

```

public static int task4CP(int[] input) {
    HashSet<Integer> numSet = new HashSet<>(); // Set to store unique elements of the array
    int maxLength = 0; // Variable to store the length of the longest consecutive sequence
    for (int num : input) {
        numSet.add(num); // Add all elements to the set to remove duplicates and order
    }
    for (int num : input) {
        if (!numSet.contains(num - 1)) { // Check if the current element is the start of a sequence
            int currentNum = num; // Initialize the current number
            int currentLength = 1; // Initialize the current sequence length
            while (numSet.contains(currentNum + 1)) { // Continue counting the consecutive sequence
                currentNum++;
                currentLength++;
            }
            maxLength = Math.max(maxLength, currentLength); // Update the maximum length if necessary
        }
    }
    return maxLength; // Return the max length
}

public static void main(String[] args) {
    int[] input = {100, 4, 200, 1, 3, 2, 5};
    int result = task4CP(input);
    System.out.println(result);
}

```